“Linux Signal-Based Process Communication and Control System”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

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Declaration

The Project Report entitled **"Linux Signal-Based Process Communication and Control System"** is a record of Bonafide work of **VENNELA S - 2520080018 , HARSHITHA B - 2520080007, KRISHNASREE M - 2520080009** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering (or) Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled **"Linux Signal-Based Process Communication and Control System"** is a bonafide work done and submitted by **KRISHNASREE M– 2520080009 ,**

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ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my Course Coordinator Dr. K. Anusha, and my guide Raghupathi**,** Department of Computer Science which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. I. Sreenath (AI&DS)**, Head of the ***Department of artificial intelligence and data structures*** for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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Abstract

The Linux Signal-Based Process Communication and Control System is designed to demonstrate how multiple processes can communicate and be controlled in a Linux environment using signals. The system consists of a controller process that creates and manages multiple worker processes. The fork() system call is used to create processes, while the kill() system call is used to send signals for controlling their activities. Worker processes use signal handlers to respond to the signals received from the controller.

The system supports process control operations such as starting, pausing, resuming, communicating with, and terminating worker processes. The controller also monitors process IDs, process status, and signal events. The wait() system call is used to synchronize with child processes and ensure their proper termination. This project demonstrates important Operating System concepts, including process management, inter-process communication (IPC), signal handling, process states, and synchronization. It provides a practical understanding of how Linux system calls are used for process control and communication.

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CHAPTER -1 INTRODUCTION

1.1 Background of the Project

An operating system manages the execution of programs and provides mechanisms for processes to communicate with one another. In Linux, a process is an instance of a running program. A system may contain multiple processes executing independently, and managing their activities is an important responsibility of the operating system.

Linux provides signals as a mechanism for notifying processes about events and requesting specific actions. Signals can be used to interrupt, pause, resume, or terminate processes, depending on the signal and the way the process handles it.

This project, Linux Signal-Based Process Communication and Control System, demonstrates the practical use of Linux signals to manage multiple processes. A controller process creates worker processes and sends them signals to control their activities. The worker processes respond to these signals through signal handlers, while the controller monitors their status and coordinates their execution.

The project helps connect theoretical concepts of operating systems with practical system programming using the C language and Linux system calls.

1.2 Problem Statement

Managing multiple processes requires a reliable mechanism for controlling their execution and coordinating their activities. Without proper process management and communication, it can become difficult to monitor process states, send control commands, and ensure that processes terminate correctly.

The objective of this project is to develop a Linux-based system that allows a controller process to create, communicate with, monitor, and control multiple worker processes using signals. It demonstrates how Linux system calls and signal-handling mechanisms can be used to manage process activities and synchronize process termination.

CHAPTER -2 SYSTEM REQUIREMENTS